

DIGESTIVE SYSTEM

Notes

1. Introduction to Anatomy & Physiology

Anatomy is the study of human body structure and the function of its parts.

Physiology is the science of body function.

Anatomy comes from the Greek word "ana-tome," meaning "cutting up."

Branches of Anatomy

- Pathological Anatomy: the study of structural changes caused by disease.
- Molecular Biology: the study of anatomical structure at the cellular level.

Anatomical Terms of Position

Anatomical terms of position and direction describe the position of one body part relative to another. They are usually based on the three major body planes.

Superior & Inferior

- Superior: a structure that is closer to the head, or higher than another structure, is said to be superior. (e.g. the human head)
- Inferior: a structure that is closer to the feet, or lower than another structure, is said to be inferior. (e.g. the human feet)

Medial & Lateral

- Medial: refers to a structure that is closer to the midline (median plane) of the body.
- Lateral: refers to a structure that is farther away from the midline than another structure.

Distal & Proximal

- Distal: refers to a structure that is farther away from the root of a limb than another structure.
- Proximal: refers to a structure that is closer to the root of a limb than another structure.

Dorsal & Ventral

- Dorsal: towards the back of the body. Example — position of the spinal cord is dorsal.
- Ventral: towards the front of the body. Example — position of the stomach is ventral.

2. Organisation of the Human Body

Anatomical structures and physiological mechanisms are arranged in a series of interacting levels of organisation — chemical, cellular, tissue, organ, and organ system.

The cell is the smallest structural and functional unit of the human body. The body is made up of millions of cells. Cells form tissue; tissue forms organs; organs form an organ system; and organ systems form the organism.

Cell → Tissue → Organ → Organ System → Organism

3. Human Digestive System — Overview

The digestive system is the collective name used to describe the alimentary canal, some accessory organs, and a variety of digestive processes.

The digestive system prepares food for use by the millions of body cells. The alimentary canal modifies food through physical and chemical modification (digestion), depending on secretion, and controls the movement of food through the digestive tract.

The digestive system is made up of the gastrointestinal tract, a tubular structure that extends from the mouth to the anus, with a length of about 30 feet, opening at both ends. All living organisms require nutrient substances and water for survival and growth, so these must be digested and absorbed by the digestive system.

Activities of the Digestive System

Ingestion → Digestion → Absorption → Assimilation → Elimination

- Ingestion: taking food into the alimentary canal — i.e., eating and drinking.
- Digestion: breakdown of food by chewing and mixing with chemical substances.
- Absorption: digested food substances pass through the alimentary canal into the blood.
- Assimilation: use of the nutrients present in the food.
- Elimination: food substances that have been eaten but cannot be digested or absorbed are excreted from the eliminatory canal as faeces, through the process of defecation.

Components of the Digestive System

Mouth → Pharynx → Oesophagus → Stomach → Small Intestine → Large Intestine

4. Mouth

The mouth (oral cavity) is bounded by muscles and bones. Food enters the digestive system through the mouth. The mouth is lined throughout with a mucous membrane. The part of the mouth between the gums and cheek is the oral cavity. The palate forms the roof of the mouth and is divided into the anterior hard palate and the posterior soft palate.

Uvula: the uvula is a curved fold of muscle, covered with mucous membrane, hanging down from the middle of the free border of the soft palate.

5. Teeth

Teeth are mineralised, horny structures projecting from the jaws. There are 20 temporary (milk) teeth, 10 in each jaw. They begin to erupt when the child is about 6 months old and should all be present by 24 months. The permanent teeth begin to replace the milk teeth from the 6th year of age, and the full set of 32 teeth is usually complete by the 24th year.

Types of Teeth

- Incisor
- Canine
- Premolar
- Molar

Incisors are the eight teeth in the front and centre of the mouth (4 on top, 4 on bottom). These are the teeth we use to take bites of food.

Canines are the sharpest teeth and are used for tearing food.

Premolars are used for chewing and grinding food.

Molars are used for chewing and grinding.

Dental Formula

2 1 2 3 (upper jaw) / 2 1 2 3 (lower jaw)

- 2 incisors
- 1 canine

- 2 premolars
- 3 molars

Structure of a Tooth

A tooth is composed of a hard, ivory-like substance called dentin, which is covered by enamel over the crown and cementum at the root. The pulp cavity contains connective tissue, blood vessels, and nerves. The blood vessels and nerves pass to the tooth through small canals in the roots.

Parts of a Tooth

- Crown
- Neck
- Root
- Apex

Crown: the part that projects from the gum and meets with one or more teeth in the opposite jaw.

Neck: the part between the crown and the root, encircling the crown.

Root: the part embedded within the socket of the jaw, below the crown.

Apex: the last part of the root.

Function of Teeth in Digestion

Mastication (chewing) is the mechanical process that breaks up large food particles into smaller pieces. Chewing is especially important for fruits and vegetables. Chewing can be carried out voluntarily, but most of it is a reflex act — the movement of the lower jaw against the upper jaw is brought about by the muscles of mastication.

Types of Dentition

Monophyodont: a dentition in which the teeth appear only once in the lifetime of the organism, placed in the socket of the jawbone.

Diphyodont: a dentition in which the teeth appear twice in the lifetime of man — the milk teeth and the permanent teeth.

Heterodont: a dentition in which the teeth are of four different types, differing both in structure and function (incisors, canines, premolars, molars).

6. Tongue

The tongue is a muscular organ situated on the floor of the oral cavity. It is attached at the posterior end and free at the front. It helps with mastication and swallowing (deglutition); it also helps in speech.

The tongue contains taste buds, which recognise four different tastes — sweet, salt, sour, and bitter. We get the sensation of taste only when moisture from food comes into contact with the taste buds. At the back of the oral cavity, on either side of the tongue, there are masses of lymphatic tissue known as tonsils.

Exocrine gland → secretes into a duct (e.g., digestive glands).

Endocrine gland → secretes directly into the blood.

Heterocrine gland → both exocrine and endocrine functions.

Salivary Reflex Pathway (as noted)

- CNS → ANS → Sympathetic / Parasympathetic
- Mechanoreceptor / Chemoreceptor stimulus → Afferent → Medulla → Efferent → Response (Unconditioned reflex)

- Conditioned reflex: Afferent → Medulla → CNS (learned association, e.g. sight/smell/sound of food)

Functions Related to the Tongue / Oral Cavity

- Lubrication → swallowing
- Digestion → carbohydrate breakdown
- Defensive → anti-pathogenic
- Protective
- Hormone → gustin

Oesophagus (Introductory Note)

Length: about 25 cm. Movement of the bolus: peristalsis. The gastro-oesophageal sphincter is made of muscle.

7. Salivary Glands and Saliva

Salivary glands are exocrine glands with ducts that produce saliva. There are 3 pairs of salivary glands: the parotid, the submandibular, and the sublingual.

Parotid Gland

These are situated on each side of the face, just below the ear. Each gland has a duct opening into the mouth (parotid duct — Stensen's duct).

Submandibular Gland

This lies on each side of the face, under the angle of the jaw. The two submandibular ducts open on the floor of the mouth (submandibular duct — Wharton's duct).

Sublingual Gland

Lies under the mucous membrane of the floor of the mouth, in front of the mandibular gland (ducts of Rivinus).

Saliva

Saliva is a watery substance located in the mouth of the organism and secreted by the salivary glands. The secretion of the salivary glands keeps the mouth and other parts of the digestive system wet and slippery. This helps food go down the throat to the stomach and act as a lubricant coat of mucus. (Volume: 1000–1500 mL; acidic in nature.)

Composition of Saliva

Saliva is composed of water (99.5%) and solids (0.5%).

- Organic solids: amylase, lipase, lysozyme, mucin/mucus
- Inorganic solids: sodium, minerals, calcium, potassium

Regulation of Salivary Secretion

Salivary secretion is under autonomic nervous control (both the parasympathetic and sympathetic nervous systems) and is regulated by reflexes.

- Unconditioned reflexes: activated by stimulation of mechanoreceptors (chewing) and chemoreceptors in the taste buds of the oral cavity. Impulses from these receptors are carried to the salivation centre in the medulla oblongata through nerves.
- Conditioned reflexes: activated by the sight, sound, or thought of food. The salivation centre also receives impulses from the brain; efferent signals are integrated and directed to the efferent part of the reflex.

Functions of Saliva

Lubrication: it helps prepare food for swallowing by lubricating it. The lubricating function of saliva allows food to pass easily from the mouth into the oesophagus. It also helps speech.

Digestion: saliva contains the enzyme amylase, which is capable of breaking down starch (CHO) into simple sugars that can be further broken down in the small intestine. Saliva also contains lipase, which plays an important role in the digestion of fat in newborn babies.

Antimicrobial action: saliva has mechanical cleansing action and helps antipathogenic action, killing pathogens in food.

Hormonal: saliva secretes gustin, which plays a role in the development of taste buds. It is also protective, protecting the oral cavity from external shock.

Role of Saliva in Taste

Saliva is very important in the sense of taste — it is the liquid medium in which chemicals are carried to the taste receptor cells.

8. Pharynx

When food is first swallowed, it passes from the mouth into the pharynx, a funnel-shaped tube that extends from the nasal cavity to the oesophagus. For descriptive purposes, the pharynx is divided into 3 parts (the Eustachian tube opens into the pharynx):

- Nasopharynx
- Oropharynx
- Laryngopharynx

Nasopharynx: the nasal part of the pharynx, situated behind the nasal cavity and above the level of the soft palate.

Oropharynx: this part extends from the level of the soft palate to the level of the epiglottis (windpipe).

Laryngopharynx: continuous with the oesophagus.

Functions

- Nasopharynx → respiration
- Oropharynx & laryngopharynx → digestion and respiration

9. Oesophagus

The oesophagus is a muscular tube about 25 cm long. It is the first part of the digestive tract that is continuous with the pharynx, and extends down to just below the stomach.

Functions

The oesophagus secretes mucus and transports food into the stomach. It does not produce digestive enzymes and does not carry out absorption.

Peristalsis

A wave-like movement that moves food to the stomach from the oesophagus.

Oesophageal Sphincter

Controls the opening of the oesophagus into the stomach.

10. Stomach

The stomach is a J-shaped enlargement of the digestive tract. It is present on the left side of the abdominal cavity. It is the muscular bag forming the widest and most distensible part of the alimentary canal. It is connected above to the lower end of the oesophagus and below to the duodenum. It acts as a reservoir of food and helps in the digestion of carbohydrate, protein, and lipids. It is about 25 cm long. The capacity of the stomach is about one ounce at birth, one litre at puberty, and 1.5 to 2 litres in an adult.

Structure of the Stomach

The stomach wall has three muscle layers: oblique fibres, circular fibres, and longitudinal fibres. Main regions: cardiac stomach, fundus, and pyloric stomach.

Parts of the Stomach

The stomach is divided into the cardiac part, fundus, and pyloric part. The oesophagus opens into the fundus/cardiac region; the body is the main part of the stomach; the pylorus is the narrow part of the stomach.

Functions

- Temporary storage, allowing time for digestive enzymes like pepsin to act.
- Mechanical breakdown of food.
- Absorption (limited).

Gastric Glands

The tubular glands found on the wall of the stomach are called gastric glands. There are about 35 million gastric glands in a human. They open into the stomach through gastric pits. The gastric glands are formed of three kinds of cells:

- Mucous neck cells: secrete mucus.
- Chief cells (zymogen cells): secrete pepsinogen.
- Oxyntic cells (parietal cells): secrete HCl and intrinsic factor.

The collective secretion of the gastric glands is called gastric juice; it is acidic in nature.

The mucous membrane of the stomach wall is thrown into ridges called rugae, which appear when the stomach is empty. The muscles of the stomach wall churn and mix the food with gastric juice, and also propel the food into the small intestine through the pyloric opening known as the pylorus.

Gastric Juice

Gastric juice is the fluid that is secreted into the stomach.

- Volume: 1200–1500 mL
- Nature: acidic
- pH: 1 to 1.5

Composition of Gastric Juice

Water: 99.5%. Solids: 0.5%, divided into organic (0.4%) and inorganic (0.1%) components.

- Organic: pepsin, mucus, gastrin, intrinsic factor (important for baby fat and protein/casein digestion during the milk-feeding period).
- Inorganic: HCl, sodium, calcium.

Function of Gastric Juice

Pepsinogen (inactive form) is converted to pepsin (active form). Pepsin is a proteolytic enzyme that helps in protein digestion; it breaks the covalent bonds between protein molecules. Pepsin works best in a strongly acidic medium (pH ~1–2), which is provided by HCl.

HCl converts pepsinogen into pepsin and kills microorganisms.

Intrinsic factor binds to vitamin B12 to promote its absorption in the small intestine.

Mucus lubricates and protects cells from the acidic chyme and pepsin.

Gastrin is a hormone that regulates stomach secretion; it is influenced by hunger.

In infants, gastric juice also contains a special proteolytic enzyme called rennin: milk protein (casein) $\xrightarrow{\text{rennin}}$ amino acids. Gastric juice also contains lipases, which initiate lipid digestion.

Regulation of Secretion of Gastric Juice

Regulation of secretion of gastric juice is produced/brought about by two mechanisms:

- Nervous regulation
- Hormonal regulation

Nervous regulation: autonomic reflexes involving impulse formation regulate gastric juice secretion.

Hormonal regulation: the hormonal mechanism is mediated by the release of the hormone gastrin. Gastrin helps regulate stomach secretion.

11. Small Intestine

The small intestine extends from the pyloric region to the large intestine. It lies in the abdominal cavity, surrounded by the large intestine, and is about 6 metres long. Its length is greater in males than in females. The small intestine is continuous with the stomach at the pyloric sphincter.

Parts of the Small Intestine

- Duodenum
- Jejunum
- Ileum

Duodenum: the shortest, widest, and most fixed part of the small intestine. It extends from the pyloric region to the beginning of the jejunum and is about 25 cm long. The 25 cm is divided into 4 parts: superior (5 cm), descending (7.5 cm), horizontal (10 cm), and ascending (2.5 cm).

Jejunum and Ileum: the jejunum constitutes 2/5 of the small intestine, while the ileum constitutes 3/5. The ileum terminates at the ileocaecal junction (the caecum is part of the large intestine).

Structure of the Villus

The villus is a finger-like projection of the intestinal wall lined by epithelial cells and microvilli, containing an arteriole, venule, and a capillary network, as well as a lacteal (lymphatic duct).

- Osmosis: movement of water from high to low concentration.
- Diffusion: movement from high to low concentration — active (uses energy), passive (without energy), and facilitated diffusion (e.g., sodium, amino acids, fructose).
- The lacteal transports absorbed fat (as chylomicrons) and protein coat; assimilation follows absorption.

Pancreatic Juice

Enzymes/secretions of the pancreas include:

- Trypsinogen $\rightarrow$ Trypsin
- Chymotrypsinogen $\rightarrow$ Chymotrypsin
- Procarboxypeptidase
- Amylase
- Lipase
- Nuclease

Trypsin, chymotrypsin, and carboxypeptidase are proteolytic enzymes: protein → dipeptide → amino acid. Glycogen → disaccharide → monosaccharide. Lipid → (lipase) → diglyceride, triglyceride, and fatty acid + glycerol. Nucleic acid → nucleoside, nucleotide, nucleoside → nitrogen base.

Note: Intestinal gland secretions include succus entericus and enterokinase.

Pancreatic juice and bile are released into the duodenum through the hepato-pancreatic duct. Pancreatic juice contains inactive enzymes — trypsinogen, chymotrypsinogen, procarboxypeptidase, amylase, lipase, and nucleases. The enzyme enterokinase, present in the intestinal juice (succus entericus), activates inactive trypsinogen into active trypsin, which in turn activates the other enzymes of the pancreatic juice.

Bile has no enzymes; it contains bile salts, bile pigments, etc. Bile helps digestion by emulsification — the conversion of fat into small micelles. It also activates lipases.

Substrate → Enzyme → Product summary:

- Proteins / Peptones / Proteoses —(trypsin, chymotrypsin, carboxypeptidase) → Dipeptides
- Polysaccharide (starch) —(amylases) → Disaccharides
- Fat —(lipases) → Diglycerides, Monoglycerides
- Nucleic acid —(nucleases) → Nucleoside, Nucleotide

The goblet cells of the intestine secrete mucus. The secretions of the mucosa and goblet cells together constitute the succus entericus, which contains maltase, dipeptidase, lipases, nucleosidases, etc. Bicarbonate and mucus provide an alkaline medium for these enzymes and also protect the intestine from acid. The intestinal juice acts on the end products of digestion, as below:

Enzymes of the Small Intestine (Intestinal Juice)

Substrate	Enzyme	Product
Dipeptides	Dipeptidase	Amino acids (A.A)
Maltose	Maltase	Glucose + Glucose
Lactose	Lactase	Glucose + Galactose
Sucrose	Sucrase	Fructose + Glucose
Nucleotide	Nucleotidase	Nucleoside
Nucleoside	Nucleosidase	Sugar + Nitrogen base
Diglyceride / Monoglyceride	Lipases	Fatty acid + Glycerol

The digested food is absorbed in the jejunum and ileum; undigested and unabsorbed substances pass on to the large intestine.

12. Large Intestine

Undigested food passes to the large intestine and is eventually eliminated as faeces, via the ileocaecal sphincter/wall. Main parts: caecum, colon (ascending, transverse, descending), rectum, and anus.

The large intestine extends from the ileocaecal junction to the anus. It is shorter and much wider than the small intestine, measuring about 1.5 m long. It shows neither villi nor brush-bordered cells, unlike the small intestine. It is differentiated into 3 regions: mainly the caecum, colon, and rectum.

The caecum is very small. In man, the caecum bears a finger-like outgrowth of unknown function, known as the vermiform appendix. Infection of this part is known as appendicitis (normal size: length 6 cm, width 7.5 cm).

The colon is an inverted U shape, with an ascending colon, transverse colon, and descending colon (pelvic colon). It is continuous with the rectum, which is concerned with the temporary storage of faeces. The rectum opens into the anus. The anus is guarded by circular, voluntary muscles called the sphincters.

The large intestine secretes no enzymes; it serves to store the unabsorbed food. It also concentrates the contents by absorbing water and forming faeces. Faeces are eliminated by the wave-like movement of the colon and rectum (defecation).

Undigested waste (faeces) enters the caecum through the ileocaecal valve, which prevents the backflow of faeces. Faeces is stored in the rectum until defaecation (elimination of faeces).

13. Absorption

The process of transfer of the end products of digestion from the intestine to the circulatory fluids, such as blood and lymph, is called absorption. It is carried out by passive, active, or facilitated transport mechanisms.

- Small amounts of glucose and amino acids are absorbed by simple diffusion.
- Some fructose and amino acids are absorbed with the help of carriers like sodium and potassium — a mechanism called facilitated transport.
- Water is absorbed by osmosis.
- Various nutrients such as amino acids and monosaccharides are absorbed actively (active transport is an energy-requiring process).
- Monoglycerides, diglycerides, and fatty acids are insoluble in water, so they cannot be absorbed directly from the intestine. With the help of bile, the end products of fat digestion are converted into small, spherical, water-soluble droplets called micelles. These are reformed into small protein-coated fat globules called chylomicrons, which are transported into the lymph (lacteal) in the villi; the chylomicrons then enter the blood via the lymph.

The absorbed material is incorporated into body tissues for their activity — this process is called assimilation. After absorption, the undigested part is eliminated (egestion) through the anus — defaecation.

Absorption in Different Parts of the Alimentary Canal

Region	Absorbs
Mouth	Certain drugs
Stomach	Water, sugar, alcohol
Small Intestine	All digested food, minerals, vitamins
Large Intestine	Water and some drugs

14. Liver

The liver is a large, solid gland situated in the upper-right and central portion of the abdomen. It is reddish-brown in colour and weighs about 1–1.5 kg in an adult man. The liver has 2 lobes (bilobed) — a large right lobe and a small left lobe. Each lobe is subdivided into numerous tiny hepatic lobules, which are the functional units. Each lobule is covered by a thin tissue called Glisson's capsule and is formed of numerous hepatic cells that radiate outward from a central vein.

The bile secreted by hepatic cells passes through the hepatic ducts and is stored in the gallbladder. The common hepato-pancreatic duct is guarded by a sphincter called the sphincter of Oddi. Bile reaches the duodenum through the bile duct, which is formed by the union of the hepatic duct and the cystic duct (duct of the gallbladder); this joins with the pancreatic duct before opening into the duodenum.

The liver secretes bile — about 800–1200 mL (up to 97.6% water, 2.4% solids). Fat is broken down through emulsification, and lipase converts fat into fatty acids and monoglycerides. The sphincter of Oddi, together with the hormones secretin and cholecystokinin, regulates the release of bile.

Composition of bile — organic: glycocholate, taurocholate, bile salts, bile pigments, fatty acids, cholesterol; inorganic: sodium, potassium, calcium (minerals). Bile pigments (bilirubin and biliverdin) come from the breakdown of red blood cells/haemoglobin.

15. Histology of the Human Gut

The wall of the alimentary canal is formed of 4 major layers, namely: Mucosa, Submucosa, Muscularis, and Serosa.

- Mucosa: the innermost layer; a moist and friable, resistant lining of epithelium that contains secretory and absorptive cells.
- Submucosa: found just outside the mucosa; a soft connective tissue layer, highly vascular and supplied with nerves.
- Muscularis externa: present external to the submucosa; contains an inner circular and an outer longitudinal muscle layer.
- Serosa: the outermost layer of the gut.

Cross-section of the gut wall, from inside out: Lumen → Mucosa → Submucosa → Muscularis (circular fibres, longitudinal fibres) → Serosa.

16. Digestion and Absorption of Food — Summary

The process of digestion is effected by both mechanical and chemical processes. The mechanical process involves mastication (chewing) and deglutition (swallowing), along with the various movements of the alimentary canal. The teeth and tongue, with the help of saliva, masticate and mix the food. Saliva helps in lubrication and in gathering the masticated food into a ball called a bolus. It is pushed backward by the tongue for deglutition.

The bolus passes through the oesophagus into the stomach by peristalsis. Saliva contains organic and inorganic substances. The chemical process of digestion begins in the buccal (oral) cavity: starch → (ptyalin) → disaccharide. Lysozyme in saliva is antipathogenic. About 30% of carbohydrate in food is digested there by salivary amylase (ptyalin).

Stomach: the mucosa of the stomach has digestive glands called gastric glands. The stomach stores food for 5 to 7 hours. Here the food is mixed with gastric juice by churning movements and converted into a pasty form known as chyme. HCl in the stomach converts inactive pepsinogen into pepsin. Protein — (pepsin) → proteoses and peptones.